

HENRY MANELSKI

Munich, Germany

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PROFESSIONAL SUMMARY

A postdoctoral researcher working with Dr. Philipp Reiß at the Technical University of Munich (TUM) with expertise in remote sensing, spectroscopy, and machine learning. Experienced member of NASA's Perseverance rover science team with strong Python and data analysis and laboratory skills.

EDUCATION

Purdue University

PhD - Planetary Science, Advisor: Roger Wiens

05 2022 – 05 2026

West Lafayette, IN

Columbia University

BSc - Applied Mathematics, GPA: 3.56

08 2018 – 05 2022

New York, NY

SELECTED COURSEWORK

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|---------------------------------------|-------------------------------|--|----------------------------------|
| • Dynamical Systems | • Mathematics of Data Science | • Laboratory Analysis of Planetary Materials | • Partial Differential Equations |
| • Geochemistry for a Habitable Planet | • Space Instrumentation | • Mars Seminar | • Space Policy (audited) |

WORK AND LEADERSHIP EXPERIENCE

SuperCam Science Team | Student Collaborator

05 2022 - 05 2026

- Worked on the Perseverance Mars rover as a science Payload Uplink Lead (sPUL) for SuperCam.
- Specialized in analysis of Laser Induced Breakdown Spectroscopy (LIBS) data, plasma diagnostics (manuscript published in 2024), and trace element geochemistry.
- Lead the effort to calibrate SuperCam to quantify nickel - which discovered one of the highest Ni enrichment ever found in bedrock on Mars. (manuscript published in 2026)

Crater Café | Lead Organizer

05 2024 - 05 2025

- Served as the lead organizer for the Purdue EAPS Planetary group's weekly seminar: Crater Café. Helping to bring world class planetary scientists to Purdue to share their research with our department.

EAPS Graduate Student Association | Secretary

09 2023 - 09 2024

- Served as a liaison for graduate students in Purdue EAPS to communicate with faculty and staff. Organized monthly meetings to plan events, discuss graduate student concerns, and make our voices heard within the department.

Jet Propulsion Laboratory | Research Assistant

06 2021 - 08 2021

- Awarded a Summer Undergraduate Research Fellowship (SURF) at JPL/Caltech for the summer of 2021. Principal component analysis and spectral parameters were used to investigate passive spectra from the Curiosity Rover's ChemCam instrument.

TEACHING EXPERIENCE

Purdue University EAPS | Teaching Assistant

01 2025 - 05 2025

- Ran lab sessions for "EAPS 112: Earth Through Time". Made custom presentations to engage students and graded lab assignments.

Columbia University Dept. of Earth & Enviro. Eng. | Teaching Assistant

01 2022 - 05 2022

- Helped to build the curriculum for and grade projects/exams for the course "EAEE 4262: Space Exploration and Mining". Held office hours and review sessions. Created lesson plans and ran discussion sessions.

Columbia University Dept. of Mathematics | Teaching Assistant

09 2020 - 12 2020

- Graded exams and held office hours for the course "MATH 1102: Calculus II". Ran review sessions before exams.

SKILLS & INTERESTS

Languages: English (Native), German (C1/B2), Mandarin (A2), Polish (A2)

Technical: Laboratory skills (lasers, optomechanics, vacuum systems), Python (Pandas, Numpy, Sci-kit learn), Multivariate regression (PLS, Elastic Net), Supervised machine learning (Random Forest, ExtraTrees), GIS (ArcGIS, QGIS, JMARS)

Citizenship: United States, Austria

AWARDS

2025 - Geology Graduate Award, given by Purdue EAPS

2025 - Cedric J. Newby Award, given by Purdue EAPS

2024 - 2nd Best Publication in Spectrochimica Acta Part B in 2024, selected for an Invited Talk at SciX 2025

2024 - Tomas Hirschfield Scholar Award at FACSS SciX